

2023-Ph-H1-Q13

Section: Particles and Waves

Topic: Wave-Particle Duality

A student makes the following statements about wave-particle duality:

- I The photoelectric effect is evidence supporting the **particle model** of light.
- II Interference is evidence supporting the **wave model** of light.
- III Photons of sufficient energy can eject electrons from the surface of metals.

Which of these statements is/are correct?

Answer: E

(I, II and III)

Guidance for Students:

This question is testing your understanding of how light exhibits both **wave** and **particle** behaviour — a central feature of **wave-particle duality**.

- Statement I is correct:

The **photoelectric effect** cannot be explained using wave theory. It requires a **quantum** explanation involving **photons** — tiny packets of energy.

- Statement II is also correct:

Interference patterns (e.g. in the double-slit experiment) are a hallmark of **wave behaviour**.

- Statement III is a factual consequence of the photoelectric effect:

A photon with sufficient energy ($E = hf$) can **eject an electron**, provided its energy exceeds the metal's **work function**.

Revision Tips:

-  **Photoelectric effect → particle model**
-  **Interference/diffraction → wave model**
-  **Dual nature = light behaves as both wave **and** particle**
- Learn:
 - $E = hf$ for photon energy
 - Work function ϕ
 - Threshold frequency concept
- Practice identifying which experiments support which model!